

The Vision Pit: Enhancing Remote Collaboration through Simulated Spatial Co-presence

Kosuke Teruyama
KYOCERA Corporation
Yokohama, Japan
kousuke.teruyama.cy@kyocera.jp

Fuko Hisada
KYOCERA Corporation
Yokohama, Japan
fuuko.hisada.ke@kyocera.jp

Jacqueline Urakami
KYOCERA Corporation
Yokohama, Japan
jacqueline.urakami.ke@kyocera.jp

Masahiro Inagaki
KYOCERA Corporation
Kyoto, Japan
masahiro.inagaki.cy@kyocera.jp

Ryuta Harima
KYOCERA Corporation
Yokohama, Japan
ryuta.harima.nf@kyocera.jp

Tomoko Hachiya
KYOCERA Corporation
Kyoto, Japan
tomoko.haba.yb@kyocera.jp

Naoyuki Tamai
Kyocera(japan)
KYOCERA Corporation
Yokohama, Japan
naoyuki.tamai.xm@kyocera.jp

Tomohiro Ogawa
KYOCERA Corporation
Yokohama, Japan
tomohiro.ogawa.yb@kyocera.jp

Abstract

This paper introduces the "Vision Pit," an innovative online meeting booth designed to enhance remote video conferencing (RVC) by replicating the spatial dynamics of in-person roundtable discussions. The Vision Pit addresses the limitations of traditional RVC in conveying non-verbal cues and fostering co-presence. It uses a curved half-mirror to facilitate natural gaze sharing among participants without requiring complex software processing and employs an audio application that generates spatial sound aligned with the participants' seating arrangement on the display. A user study comparing the Vision Pit with standard RVC revealed that participants experienced increased co-presence, improved use of non-verbal communication, and a more balanced participation rate. The Vision Pit offers a promising, cost-effective solution for enriching remote communication by prioritizing natural interaction and shared spatial awareness.

CCS Concepts

• **Human-centered computing** → Collaborative and social computing; Collaborative and social computing devices; Collaborative and social computing; Empirical studies in collaborative and social computing.

Keywords

Remote video conferencing, Co-presence, non-verbal communication, communication

ACM Reference Format:

Kosuke Teruyama, Fuko Hisada, Jacqueline Urakami, Masahiro Inagaki, Ryuta Harima, Tomoko Hachiya, Naoyuki Tamai, and Tomohiro Ogawa. 2025. The Vision Pit: Enhancing Remote Collaboration through Simulated Spatial Co-presence. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3706599.3719840>

1 Introduction

In response to the widespread adoption of remote work environments and online meetings due to the COVID-19 pandemic, the use of remote video conferencing (RVC) systems has become commonplace. However, these systems often fail to convey essential non-verbal cues, such as eye contact, mutual gaze, and shared spatial awareness, which are crucial for effective communication in face-to-face interactions [3, 13, 18]. This limitation arises from their two-dimensional layout and the absence of spatial co-presence [2, 23]. To address these challenges, we have developed an innovative online meeting booth, termed the "Vision Pit," which replicates real-world spatial settings to enhance the use of non-verbal cues during discussions. A distinctive feature of the Vision Pit is its ability to simulate the experience of participants sitting around a round table. This configuration contrasts with the typical two-dimensional arrangement in online meetings, where participants appear side-by-side on a screen. The Vision Pit employs a curved half-mirror to facilitate gaze sharing among participants. Drawing inspiration from teleprompters, it integrates a large display, a curved half-mirror, and web cameras to achieve gaze alignment with multiple conversation partners without the need for software processing. Unlike traditional teleprompters that use a flat half-mirror, our use of a curved mirror expands the field of view, enabling gaze alignment among participants. Moreover, an application that simulates spatial sound is used to enhance the sense of spatial co-presence, an effect that has been reported in previous studies [22, 28]. Our

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3719840>

system is both simple and cost-effective. It relies on readily available hardware components, such as large displays, web cameras, and half-mirrors, making it affordable to construct. Furthermore, since it does not require high-load applications like AI-powered gaze alignment software, it can be operated on a standard PC. This novel approach offers a practical and accessible solution to enhance communication in remote meeting environments. We conducted a user study to evaluate the Vision Pit, comparing it with traditional RVC. Participants reported a greater sense of co-presence and improved use of non-verbal cues. Conversation analysis revealed a more balanced participation rate, with contributions from all participants.

2 Related Work

Non-verbal cues, which include any form of communication that does not involve words, are a crucial aspect of human interaction generally [6, 10, 15], and are equally important in HCI [7, 19, 26]. These cues are diverse, including gestures, body and eye movements, facial expressions, and spatial distances, to name a few. Non-verbal signals such as head nods, eye contact, and hand gestures play a vital role in managing and guiding the flow of conversation and turn-taking between speakers [4, 14, 27]. Eye behavior is essential for successful social interactions. Eye contact can either facilitate or obstruct communication [4, 14, 27]. Mutual gaze typically serves as the initial step in initiating an interaction, indicating awareness of the other person and a willingness to engage [1]. The absence of such cues can complicate conversations in larger groups, a problem frequently noted in remote video conferences (RVC), where the flow of conversation is disrupted due to the lack of shared gaze attention that is essential for providing feedback, regulating conversation flow, and fostering social connections [9, 25]. To address this problem, [16] incorporated web-based gaze detection and visualization in an online meeting platform, discovering that the inclusion of gaze attention enriched the interactive experience and increased user engagement in online meetings. The system by [17] has particularly focused on improving gaze cues through eye-based joint attention support by providing real-time and retrospective eye gaze information to support the establishment, maintenance, and reflection of joint attention during remote team meetings. [8] employed a gaze cursor to monitor attention and coordinate awareness in remote collaboration. The shared gaze cursor was especially beneficial when processing complex information, clarifying confusion between participants, and facilitating agreement.

Another challenge in RVC is the lack of shared spatial awareness and co-presence. [12] compared face-to-face, audio-only, and video interactions, finding that although video improved understanding and response forecasting and provided certain non-verbal information, it still fell short in managing difficult interactions, side conversations, or controlling the conversation flow. The authors concluded that information about the shared space needs to be integrated into RVC for improvement. [24] transformed participants' video streams into 3D portraits, and user feedback suggested improved communication efficacy and engagement. Similarly, [5] developed a virtual 3D meeting space to enhance shared spatial awareness. The 3D environment was more beneficial to capturing

attention and creating a sense of co-presence. However, the somewhat unnatural appearance of participants in the 3D design made it feel less comfortable and professional, emphasizing the importance of creating a natural 3D environment to avoid awkwardness in remote conversations.

3 Conceptual design of the Vision Pit

The Vision Pit is a box-shaped meeting booth designed to facilitate virtual round table discussions (see Figure 1). It features a desk and chair for a single user to conduct PC work, along with headphones and lighting specifically for meetings. A reverse conical half-mirror surrounds the user. Positioned behind this half-mirror, and out of the user's sight, are web cameras used for the meeting. The display for video conferencing is mounted on the ceiling, allowing the user to view the meeting's visuals as they are reflected in the mirror. By wearing headphones, the user experiences the voices of conversation partners as if they are seated around a round table, providing a three-dimensional auditory experience. The booth's exterior is covered with sound-absorbing materials and mesh fabric, making it easy to install in offices and other locations.

The following example explains how the prototype of the Vision Pit operates. Consider a scenario with four participants, each utilizing a Vision Pit labeled A, B, C, and D (see Figure 2 left). The prototype establishes six simultaneous one-on-one online meeting connections (A&B, B&C, C&D, D&A, A&C, B&D) through a custom application. The video captures of each participant are spatially arranged in a virtual round table as displayed in Figure 2 (middle and right).

Another application is employed to perform a polar coordinate transformation on the browser screen in real-time, which is then displayed on the ceiling-mounted display. This transformed screen is reflected in the reverse conical half-mirror of the main unit, allowing the participant to perceive a virtual image that accurately replicates spatial relationships (see Figure 3). The image of the round table is incorporated via software. Three web cameras are positioned around the user behind the half-mirror, enabling the user to naturally align their gaze with others without being aware of the cameras. The audio setup uses an application called Binaural Meet [11] that replicates spatial sound, allowing voices to be heard in three dimensions matching the spatial layout of participants displayed.

4 Method

4.1 Participants

The study involved a total of sixteen participants, including four females, aged between 20 and 50 years. Ethical approval for the study was received from the ethics committee of the Institute of Science Tokyo. Participation was voluntary, and informed consent was obtained from all participants prior to their involvement in the study. Participants were divided into four groups, balanced by gender (one female and three males) and age. They were either only vaguely acquainted with each other or were meeting for the first time.

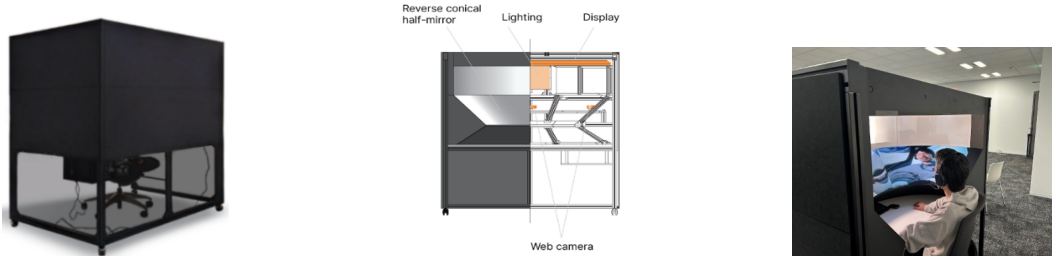


Figure 1: Exterior design of the Vision Pit (left); Assembly Drawing (middle); User perspective (right).



Figure 2: System diagram (left), real-time polar coordinate transformation (middle), participants view (right).

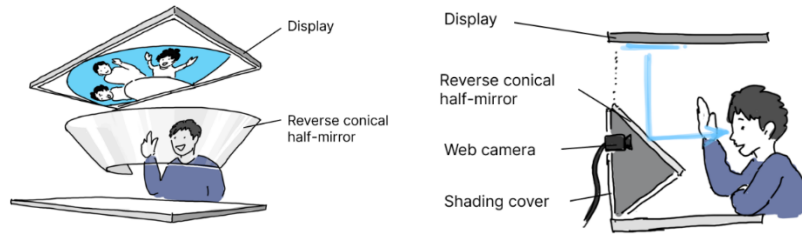


Figure 3: Ceiling-mounted display and conical half mirror of the Vision Pit.

4.2 Experimental Design and Procedure

Group discussions with RVC and the Vision Pit were conducted using a within-subject design, with RVC and Vision Pit conditions and discussion topics (see Appendix A1) counterbalanced. The RVC discussion was carried out using a PC with headset. The Vision Pit discussion was conducted with the setup described in section 3. Prior to the discussions with RVC and the Vision Pit, participants completed one discussion round face-to-face with discussion topic 1. After each session, participants filled out the Discussion Evaluation Questionnaire. Electrodermal activity was measured using Biosignalsplux (<https://www.pluxbiosignals.com>) sensors on the left-hand pointer and middle finger of one participant in each group. Discussions were recorded and analyzed with Hylable software (<https://www.hylable.com>).

4.3 Materials

4.3.1 Discussion Evaluation Questionnaire. The questionnaire contained 26 items rated on a 5-point Likert scale ranging from 1 ("I don't agree") to 5 ("I agree very much"). These items were grouped into seven categories: 1) Nonverbal signals, 2) Co-presence, 3)

Understanding agreement/disagreement, 4) Speech timing, 5) Generating opinions, 6) Discussion effectiveness, and 7) Discussion smoothness (refer to Appendix A2 for details). The questionnaire was adapted from [5] and translated to Japanese.

4.3.2 Experimental Task. In this experiment, two discussion topics were carefully chosen based on participants' familiarity and assessed for suitability through a pretest. The discussion process was structured into two distinct phases. Phase 1 focused on divergent thinking, where participants were encouraged to generate creative ideas and explore various possible solutions. This phase lasted for 5 minutes. Phase 2 transitioned into convergent thinking, where the participants worked together to decide which idea should be implemented. After the initial 5-minute phase, the experimenter introduced additional information that imposed a specific condition on the option to be selected. The discussion format underwent pretesting, revealing that it facilitated coordination and interaction while maintaining engagement within a 10-minute time frame. The introduction of new information at the midpoint revitalized the discussion, prompting participants to reassess their original ideas

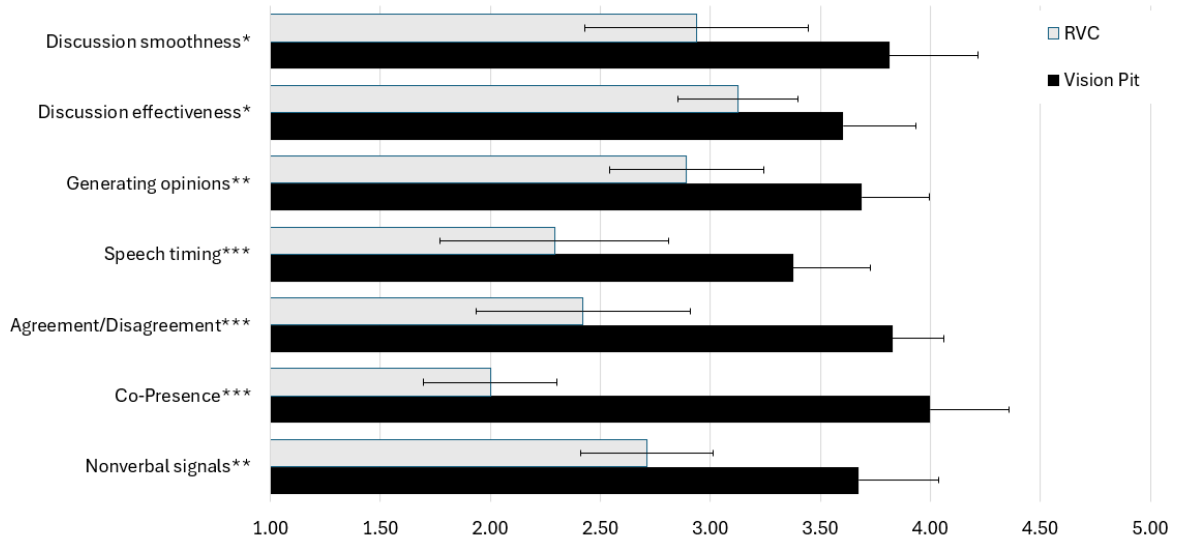


Figure 4: Participants' evaluation of RVC and Vision Pit, * $p < .05$; ** $p < .01$; *** $p < .001$

and reach compromises where necessary. Appendix A1 describes the discussion topics and additional information given.

5 Results

All data were tested for normal distribution using the Kolmogorov Smirnov test and when violations of assumptions were found a non-parametric test was applied.

5.1 Questionnaire

Overall, the Vision Pit received a better evaluation than RVC in all seven categories: 1) Nonverbal signals ($t(15) = 3.97, p < .01$), 2) Co-presence ($t(15) = 7.02, p < .001$), 3) Agreement/disagreement ($t(15) = 5.65, p < .001$), 4) Speech timing ($t(15) = 3.16, p < .01$), 5) Generating opinions ($t(15) = 3.37, p < .01$), 6) Discussion effectiveness ($t(15) = 2.23, p < .04$), and 7) Discussion smoothness, ($t(15) = 2.43, p < .03$). Participants reported that the Vision Pit was more effective in using nonverbal signals such as facial expressions, hand and head movements, was more effective in using eye contact when communicating with others, was easier to understand whether others agreed or disagreed, was easier to coordinate speech timing, was easier to generate opinions and more fun to talk, was easier to share ideas and reach consensus, and discussion were smoother and less stressful (see Figure 4).

5.2 Communication Patterns

The audio recordings from each discussion were analyzed using Hylable, a conversation analysis software that evaluates communication patterns through acoustic signals, such as speaking time for each participant [29]. Unlike text-based analysis, this software focuses on the temporal patterns of speech events for each speaker. The discussions were examined using a moving window of one second. Three key aspects of communication were assessed: 1) Activity, 2) Turn-taking, and 3) Speaker overlap. Figure 5 presents a detailed analysis of the speaking time for each participant in both

the Vision Pit and RVC conditions. Each participant is represented by a distinct color, and the graphs are divided into the first and second phases of the discussion. The graphs show that discussions were notably more active in the second phase for the Vision Pit compared to RVC, indicating an active exchange of ideas and the involvement of all participants in the decision-making process. The distribution of activity across participants, as indicated by the different colors, also suggests a more balanced participation rate among participants in the Vision Pit condition compared to the RVC condition. Overall, the speech analysis suggests that discussions in the Vision Pit condition resulted in a more balanced speech rate among participants and across time. This was particularly noticeable for participants who had a lower activity rate during the RVC discussions.

Speakers' activity is defined by the total speaking time during the discussion. Speaking time is analyzed using a moving window of one second. Each speaker is assigned a value between 0 (has not been speaking) and 1 (been speaking continuously) for each window, and these values are then summarized over the entire session. Figure 6 shows the speaking activity for each speaker of each team with the Vision Pit and RVC. Speaking activity did not differ between Vision Pit ($M = 43.82, SD = 17.02$) and RVC ($M = 37.08, SD = 18.33$); $t(30) = 1.08, p = .29$. However, the radar graphs in Figure 6 indicate that speaking time was more evenly distributed among speakers with the Vision Pit than with RVC. Notably, participants who contributed less during the RVC sessions appeared to engage more actively in the Vision Pit discussions.

Turn-taking is measured by counting the number of times a speaker becomes the leading speaker, based on moving windows of five seconds. No difference in the number of turns taken was found between Vision Pit and RVC conditions, $t(30) < 1, p > .05$. In both conditions turn-taking occurred around 6 times during the 10 minutes discussion (Vision Pit: $M = 5.81, SD = 2.90$; RVC: $M = 6, SD = 3.1$).

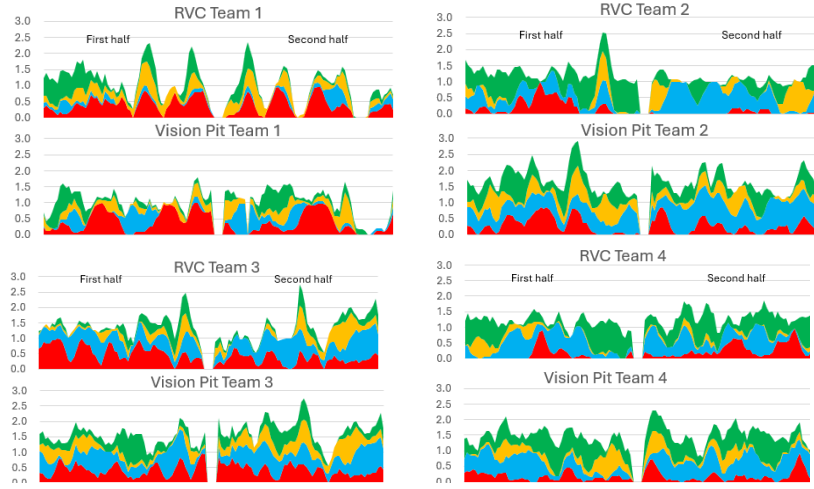


Figure 5: Communication patterns for each team in RVC and Vision Pit Session.

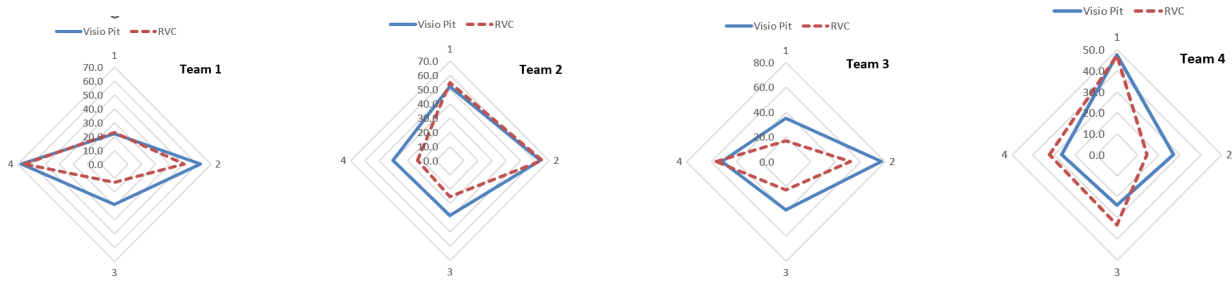


Figure 6: Radar graphs of the speaking activity of team members for each team in Vision Pit and RVC.

Speaker overlap is evaluated by analyzing the speaking time of an individual when they are not the leading speaker. Unlike speaking activity, speaker overlap excludes the leading speaker's speaking time. It is calculated using five-second moving windows and serves as an indication of how engaging and active the discussion was among group members. The analysis revealed a higher speaker overlap in the Vision Pit condition ($M = 21.94$, $SD = 6.91$) compared to the RVC condition ($M = 16.13$, $SD = 4.87$), $t(30) = 2.75$, $p = .01$. Participants speaking activity correlated positively with the evaluation of using non-verbal codes effectively ($r = .34$) and the ability to voice one's opinion ($r = .41$). Furthermore, positive relationships were found between the speaker's overlap and the evaluation of using non-verbal codes ($r = .50$), the feeling of co-presence ($r = .37$), speech timing ($r = .44$), the ability to voice one's opinion ($r = .40$) and the effectiveness of the discussion ($r = .35$). No relationship was found between turn taking and questionnaire variables. All reported correlations were significant for a p-value of $p < .05$.

5.3 Electrodermal Activity (EDA)

Electrodermal activity was measured for one participant from each group. Figure 7 illustrates the differences in Skin Conductance Level (SCL) for each participant across the Face-to-Face, Vision

Pit, and RVC conditions. Data for the Face-to-Face session are included here to serve as a baseline for comparison with Vision Pit and RVC. Due to the small sample size, statistical analyses were not conducted, and the data were interpreted qualitatively. For three out of four participants, SCL values appeared to be lowest in the Face-to-Face condition. SCL levels were high for all four participants in the RVC condition, while for two participants, SCL levels were noticeably lower in the Vision Pit condition compared to the RVC condition. In general, higher SCL levels indicate an increase in arousal or higher cognitive load [20, 21]. Results indicate that the Face-to-Face condition presented a situation where participants could relax most while the RVC condition was overall more intense for participants.

6 Discussion

Participants evaluation of the discussion and an analysis of communication patterns indicated that the Vision Pit effectively facilitated the use of non-verbal signals, fostering a sense of co-presence and spatial awareness aligning with research emphasizing the crucial role of these cues for effective interactions [26]. The improved sense of co-presence agrees with findings by [5] and [24], who demonstrated the benefits of 3D environments and avatars in enhancing

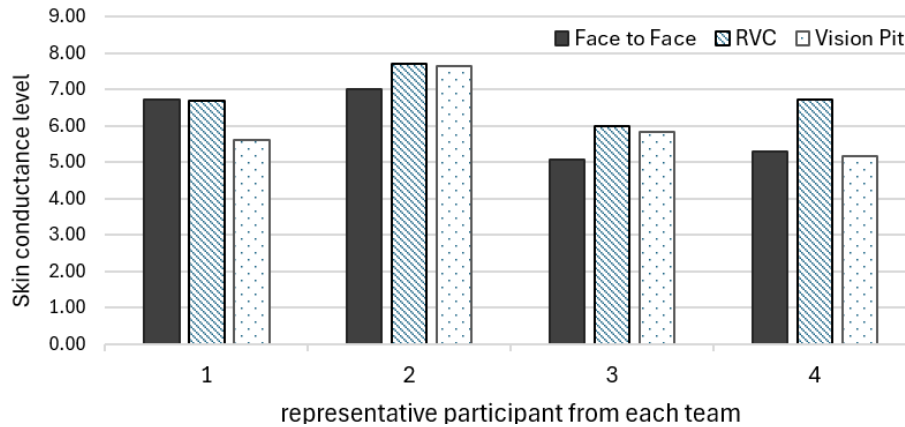


Figure 7: Skin conductance level for one participant per group in Face-to-Face, RVC, and Vision Pit.

shared spatial awareness and engagement. Participants evaluated the discussions within the Vision Pit as being smoother and less stressful, a finding supported by the lower skin conductance levels observed in some participants. The results suggest that the layout of the Vision Pit created a natural-feeling space that supported the use and perception of non-verbal cues during discussions unlike the somewhat artificial 3D setting by [5]. The enhanced perception of non-verbal cues and co-presence likely contributed to the better management of the conversation flow, addressing the limitations of traditional RVC highlighted in [12].

The increased speaker overlaps in the Vision Pit, combined with positive subjective ratings of discussion smoothness, presents an interesting finding. While traditionally considered disruptive, the overlap in this context seems to have played a positive role, possibly reflecting a more dynamic and collaborative exchange with improved speech timing, where participants felt comfortable interjecting and building on each other's ideas.

The study's limitations include a relatively small sample size, particularly concerning SCL measurements. The communication pattern analysis focused solely on the presence of speech, without examining its content. Future analyses need to consider the number and originality of ideas discussed, as well as the appropriateness and agreement on the final decision. Additionally, non-verbal behaviors such as gestures and head movements will be analyzed in future steps to complement the communication analysis.

7 Conclusion

This work introduces the Vision Pit, a novel approach to enhancing remote video conferencing by recreating the spatial dynamics of in-person roundtable discussions using a curved mirror and spatial sound. The Vision Pit uses readily available hardware and has a simple yet effective design. In a user study we could show that the Vision Pit enhances a sense of co-presence and facilitates the use of nonverbal cues, leading to more balanced participation, compared to traditional RVC. Our findings demonstrate the potential of spatially-aware interfaces to moderate the communicative limitations of existing video conferencing platforms.

Acknowledgments

We thank all participants who volunteered for the study. We would also like to express our appreciation for the support we have received from our co-workers from Kyocera such as, Hiroko Asabu for her thoughtful guidance and continuous support throughout the project, Naoya Wada (Ph.D.), for his support in the spatial audio conferencing software, and we are also deeply grateful to Masashi Fujisawa, Satoko Omura, Yasutaka Kaneda, Wakana Nakai, and Masafumi Maruo for their dedicated efforts in creating the conference and image conversion applications. Their contributions were essential to this research.

References

- [1] Michael Argyle, Mark Cook, and Duncan Cramer. 1994. Gaze and mutual gaze. *The British Journal of Psychiatry* 165, 6: 848–850.
- [2] Allison Bayro, Yalda Ghasemi, and Heejin Jeong. 2022. Subjective and objective analyses of collaboration and co-presence in a virtual reality remote environment. *2022 IEEE conference on virtual reality and 3d user interfaces abstracts and workshops (vrw)*, IEEE, 485–487.
- [3] Leanne S Bohannon, Andrew M Herbert, Jeff B Pelz, and Esa M Rantanen. 2013. Eye contact and video-mediated communication: A review. *Displays* 34, 2: 177–185.
- [4] Roser Cañigueral and Antonia F de C Hamilton. 2019. The role of eye gaze during natural social interactions in typical and autistic people. *Frontiers in psychology* 10: 560.
- [5] Thomas Joseph Cashman, Tim Hutton, Martin de La Gorce, et al. 2024. An Equal Seat at the Table: Exploring Videoconferencing with Shared Spatial Context combined with 3D Video Representations. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 1–9.
- [6] Justine Cassell, Yukiko I Nakano, Timothy W Bickmore, Candace L Sidner, and Charles Rich. 2001. Non-verbal cues for discourse structure. *Proceedings of the 39th Annual Meeting of the Association for Computational Linguistics*, 114–123.
- [7] Irina Cristescu. 2008. Emotions in human-computer interaction: the role of nonverbal behavior in interactive systems. *Revista Informatica Economică nr 2*, 46: 110–116.
- [8] Sarah D'Angelo and Darren Gergle. 2016. Gazed and confused: Understanding and designing shared gaze for remote collaboration. *Proceedings of the 2016 chi conference on human factors in computing systems*, 2492–2496.
- [9] Sarah D'angelo and Bertrand Schneider. 2021. Shared gaze visualizations in collaborative interactions: Past, present and future. *Interacting with Computers* 33, 2: 115–133.
- [10] Judith A Hall, Terrence G Horgan, and Nora A Murphy. 2019. Nonverbal communication. *Annual review of psychology* 70, 1: 271–294.
- [11] Shoichi Hasegawa. 2022. 近接チャットと HMD VR によるメタバースの相違点. [Differences between proximity chat and metaverse with HMD VR]. パーチャル学会運営委員会 [Virtual Conference 2022].
- [12] Ellen A Isaacs and John C Tang. 1993. What video can and can't do for collaboration: a case study. *Proceedings of the first ACM International Conference on*

- Multimedia*, 199–206.
- [13] Allison Jing, Kieran May, Brandon Matthews, Gun Lee, and Mark Billingham. 2022. The impact of sharing gaze behaviours in collaborative mixed reality. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2: 1–27.
 - [14] Hiromi Kobayashi and Shiro Kohshima. 2001. Evolution of the human eye as a device for communication. *Primate origins of human cognition and behavior*: 383–401.
 - [15] Igor Kotlyar and Dan Ariely. 2013. The effect of nonverbal cues on relationship formation. *Computers in Human Behavior* 29, 3: 544–551.
 - [16] Chandan Kumar, Bhupender Kumar Saini, and Steffen Staab. 2024. Enhancing Online Meeting Experience through Shared Gaze-Attention. *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, 1–6.
 - [17] Moritz Langner, Peyman Toreini, and Alexander Maedche. 2022. Eyemeet: A joint attention support system for remote meetings. *CHI Conference on Human Factors in Computing Systems Extended Abstracts*, 1–7.
 - [18] Svea Luebstorf, Joseph A Allen, Emilee Eden, William S Kramer, Roni Reiter-Palmon, and Nale Lehmann-Willenbrock. 2023. Digging into “zoom fatigue”: A qualitative exploration of remote work challenges and virtual meeting stressors. *Merits* 3, 1: 151–166.
 - [19] Divine Maloney, Guo Freeman, and Donghee Yvette Wohn. 2020. “Talking without a Voice” Understanding Non-verbal Communication in Social Virtual Reality. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW2: 1–25.
 - [20] Nargess Nourbakhsh, Fang Chen, Yang Wang, and Rafael A Calvo. 2017. Detecting users’ cognitive load by galvanic skin response with affective interference. *ACM Transactions on Interactive Intelligent Systems (TiiS)* 7, 3: 1–20.
 - [21] Nargess Nourbakhsh, Yang Wang, Fang Chen, and Rafael A Calvo. 2012. Using galvanic skin response for cognitive load measurement in arithmetic and reading tasks. *Proceedings of the 24th Australian computer-human interaction conference*, 420–423.
 - [22] Kate Nowak, Lev Tankelevitch, John Tang, and Sean Rintel. 2023. Hear We Are: Spatial Audio Benefits Perceptions of Turn-Taking and Social Presence in Video Meetings. *Proceedings of the 2nd Annual Meeting of the Symposium on Human-Computer Interaction for Work*, 1–10.
 - [23] Nanda Amalia Putri, Martina Piranti, and Dian Ariani. 2023. Exploring Co-Presence in Virtual Work: A Literature Review. *PERSPEKTIF* 12, 4: 1261–1273.
 - [24] Xun Qian, Feitong Tan, Yinda Zhang, et al. 2024. ChatDirector: Enhancing Video Conferencing with Space-Aware Scene Rendering and Speech-Driven Layout Transition. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–16.
 - [25] Martin Schuessler, Luca Hormann, Raimund Dachselt, Andrew Blake, and Carsten Rother. 2024. Gazing Heads: Investigating Gaze Perception in Video-Mediated Communication. *ACM Transactions on Computer-Human Interaction* 31, 3: 1–31.
 - [26] Jacqueline Urakami and Katie Seaborn. 2023. Nonverbal cues in human-robot interaction: A communication studies perspective. *ACM Transactions on Human-Robot Interaction* 12, 2: 1–21.
 - [27] Boris Velichkovsky, Marc Pomplun, and Johannes Rieser. 1996. Attention and communication: Eye-movement-based research paradigms. In *Advances in psychology*. Elsevier, 125–154.
 - [28] Matthew Wong and Ramani Duraiswami. 2021. Shared-space: Spatial audio and video layouts for videoconferencing in a virtual room. *2021 Immersive and 3D Audio: from Architecture to Automotive (I3DA)*, IEEE, 1–6.
 - [29] Mizomoto Takeshi and Kohei Yagira. 2022. 対面とオンラインの会話可視化技術とその EdTech への応用 [Conversation visualization technology for face-to-face and online settings and its application to EdTech]. *研究報告音声言語情報処理 (SLP)* 2022, 13: 1–5.

A APPENDICES

A1. Discussion Topics

Theme 1: Lunch Meeting

You are colleagues in a department at [company name]. Next week, a guest from another company location will visit, and you will all go out for lunch together. He is the head of the department, but he is very friendly and seems to prefer a casual lunch. Please discuss where would be a good place to take him and decide together. For example, there are places like Chinatown, Ootoya, and Ippudo. While considering other options, discuss the characteristics, advantages, concerns, and any specific restaurants you have in mind.

Information added after 5 minutes of discussion: The person is German and a vegetarian.

Thema 1 -Japanese original

テ--マ1:ランチ会

みなさんは京セラのみなとみらいリサ--チセンタ--のある部署の同僚です。来週、京セラの他の拠点からゲストが来社され、みんなで一緒にランチに行くことになりました。

彼は部の責任者ですがとてもフレンドリ--で、カジュアルなランチを希望しているようです。連れていくとしたらどこがいいか、みんなで相談して決めてください。

たとえば、中華街 大戸屋 一風堂などありますね。

そのほかの選択肢も考えながら、どういう特徴があるか、良い点、懸念点、店の心当たり、など話してみてください。

(後半)【後出し情報】 その人はドイツ人でベジタリアンです。

Theme 2: Summer Recreation

You are colleagues in a department at [company name]. Next month, there will be a summer recreation (team building) within the department, and you have been tasked with planning it. Please discuss and decide together. For example, there are options like cruising in Hayama, watermelon splitting, and a Yakatabune boat ride. While considering other options, discuss the characteristics, advantages, and concerns of each.

Information added after 5 minutes of discussion: There is a new female employee who uses a wheelchair.

Thema 2 - Japanese original

テ--マ~~2~~:夏レク

みなさんは京セラのみなとみらいリサ--チセンタ--のある部署の同僚です。

来月、部署内で夏レク(チ--ムビルディング)をする事になり、企画をまかされました。みんなで相談して決めてください。

たとえば 葉山でクル--ジング スイカ割り 屋形船などがありますね。

そのほかの選択肢も考えながら、それぞれどういう特徴があるか、良い点、懸念点、など話してみてください。

(後半)【後出し情報】 新入社員で車椅子の女性がいます。

Theme 3: Study Session

You are colleagues in a department at [company name]. You regularly hold study sessions within the department. Please discuss and decide what the next topic should be. For example, there are topics like an introduction to psychology, technology trends, and trends in social media among young people. While considering other options, discuss the characteristics, advantages, and concerns.

Information added after 5 minutes of discussion: There is a budget, so it is possible to invite an external lecturer.

Thema 3 - Japanese original

テ--マ~~3~~:勉強会

みなさんは京セラのみなとみらいリサ--チセンタ--のある部署の同僚です。定期的に部署内で勉強会を開催しています。

次回のネタは何かいいか、相談して決めてください。

たとえば心理学入門 テクノロジ--トレンド 若者のSNSトレンドなどがありますね。

そのほかの選択肢も考えながら、どういう特徴があるか、良い点、懸念点、など話してみてください。

(後半)【後出し情報】 ある程度予算が使えるので、外部から講師を招くこともできるそうです。

A2. Discussion evaluation

1. Nonverbal signals

- I felt that I could convey my intentions non-verbally.
- 言葉以外でも(非言語で)自分の意思を伝えられると感じた。
- I sensed that people other than myself were communicating non-verbally.
- 自分以外の人どうしが非言語で意思疎通していると感じた。
- I noticed that my hands or head were moving unconsciously.
- 気が付くと手や頭が動いていた。
- I felt that my facial expressions became more animated while speaking.
- 話している時に表情が豊かになっていてと感じた。
- I felt there were no ways to convey my intentions other than words.
- 言葉以外に自分の意思を伝える方法が無いと感じた。

2. Co-presence

- I felt that participants were sharing the same space.
- 参加者が同じ空間を共有していると感じた。
- I sensed that the listener was looking at the speaker while listening.
- 聞き手が話している人の方を見て聞いていると感じた。
- I felt that the speaker was looking at the listener while speaking.
- 話している人が聞き手の方を見て話していると感じた。
- It was difficult to tell who the speaker and listener were looking at.
- 話し手と聞き手が誰を見ているか、わかりにくかった。

3. Understanding agreement/disagreement

- I received signs of agreement or disagreement from someone while I was speaking.
- 自分が話している時に、誰かから同意や不同意のサインを受け取った。
- I sent signs of agreement or disagreement while someone else was speaking.
- 誰かが話している時に、自分から同意や不同意のサインを送った。
- People other than the speaker were exchanging signs of agreement or disagreement.
- 話している以外の人同士で同意や不同意のサインを送り合っていた。
- It was difficult to send or receive signs of agreement or disagreement.
- 同意や不同意のサインを送ったり受けたりし難かった。

4. Speech timing

- I found it easy to grasp the timing to start speaking.
- 自分が話し出すタイミングを掴み易いと感じた。
- I could sense when the next speaker wanted to take over.
- 次の話者が交代したがっているのを感じることができた。
- It was difficult to grasp the timing to start speaking or to hand over to someone else.
- 自分が話し出すタイミングや、誰かに交代するタイミングが掴みにくかった。

5. Generating opinions

- I felt that free opinions were easy to express.
- 自由な意見が生まれやすいと感じた。
- I felt that new ideas were sparked by others' opinions.
- 相手の意見に刺激されて新しいアイデアが生まれると感じた。
- I found it enjoyable to speak my thoughts.
- 自分の思い付きを話すのが楽しいと感じた。
- I felt that free opinions and creative ideas were hard to express.
- 自由な意見や創造的なアイデアは出にくいと感じた。

6. Discussion effectiveness

- I felt that it was easy to reach a consensus.
- 意見がまとまり易いと感じた。
- I felt that it was easy to share ideas.
- 考えを共有し易いと感じた。
- I felt that time was easily wasted on unfocused and aimless talk.
- 散漫で無駄ばなしに時間を浪費し易いと感じた。

7. Discussion smoothness

- I felt that it was a meeting that met my expectations.
- 自分の期待に応える会議だったと感じた。
- I felt that it was a meeting where it was easy to express opinions, and the discussion was smooth.
- お互いの意見を言いやすく、スムーズな会議だと感じた。
- I felt that it was a meeting with a sense of intimidation or stress.
- 威圧感やストレスを感じる会議だと感じた。